


Class list :-

- ① add
 - ② remove
 - ③ toggle
 - ④ contains
- 4 properties of classList

In JavaScript, the classList property is a read-only property of an element that returns a DOMTokenList collection of the classes assigned to the element. It provides a convenient way to manipulate the class attribute of an element.

Hamne style.css mei .class1, .class2, .class3 ko style dedi But hamari html mei aisi koi classes hai hi nahi. Now we will create 3 classes using JS code.

```
<body>
```

```
<h1>Heading 1</h1>
```

```
<h2>Heading 2</h2>
```

```
<h3>Heading 3</h3>
```

```
<script src="app.js"></script>
</body>
```

→ index.html

```
.class1{
  color: red;
}
```

```
.class2{
  font-size: 70px;
}
```

```
.class3{
  border: 10px solid black;
}
```

→ style.css

① add :-

```
let h1 = document.querySelector('h1');  
let h2 = document.querySelector('h2');  
let h3 = document.querySelector('h3');
```

→ app.js

```
h1.classList.add('class1');
```

```
h2.classList.add('class2');
```

```
h3.classList.add('class3');
```

Heading 1

Heading 2

Heading 3

→ Here we applied css properties of class 1, class 2, class 3 to h1, h2, h3.

```
let h1 = document.querySelector('h1');  
let h2 = document.querySelector('h2');  
let h3 = document.querySelector('h3');
```

```
h1.classList.add('class1');
```

```
h1.classList.add('class2');
```

```
h2.classList.add('class2');
```

```
h3.classList.add('class3');
```

→ this will also work fine.

→ h1 ko two classes dedi yaha.

Heading 1

Heading 2

Heading 3

```
h1.classList.add('class1', 'class2', 'class3');
```

→ this line will work too.

② remove :-

```
let h1 = document.querySelector('h1');  
let h2 = document.querySelector('h2');  
let h3 = document.querySelector('h3');
```

```
h1.classList.add('class1', 'class2', 'class3');
```

```
h2.classList.add('class2');
```

```
h3.classList.add('class3');
```

```
h1.classList.remove('class1', 'class3');
```

→ h1 par se
class1, class3
ki properties
hata di

Heading 1

Heading 2

Heading 3

③ contains() → ye true / false dega s ye batayega
ki particular element par ye
wali class hai ya nahi.


```

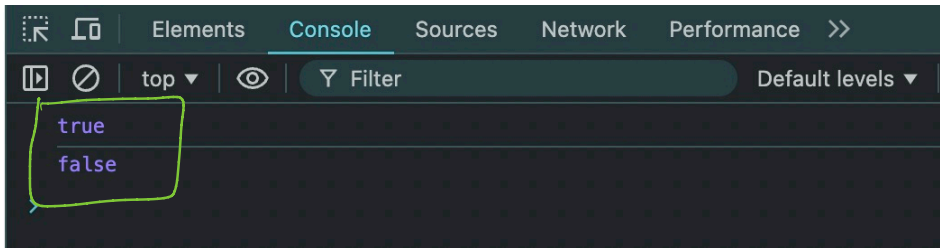
let h1 = document.querySelector('h1');
let h2 = document.querySelector('h2');
let h3 = document.querySelector('h3');

h1.classList.add('class1', 'class2', 'class3');

console.log(h1.classList.contains('class2'));
console.log(h3.classList.contains('class3'));

```

yes, h1 mei
class 2 applied
hai
no, h3 mei class 3
nahi
hai.



9

```

let h1 = document.querySelector('h1');
let h2 = document.querySelector('h2');
let h3 = document.querySelector('h3');

h1.classList.add('class1', 'class2', 'class3');

h2.classList.add('class2');

h3.classList.add('class3');

h3.classList.toggle('class3'); → 11

```

Toggle ⇒ enable / disable OR on / off

→ class 3 hatt gaya h3 se

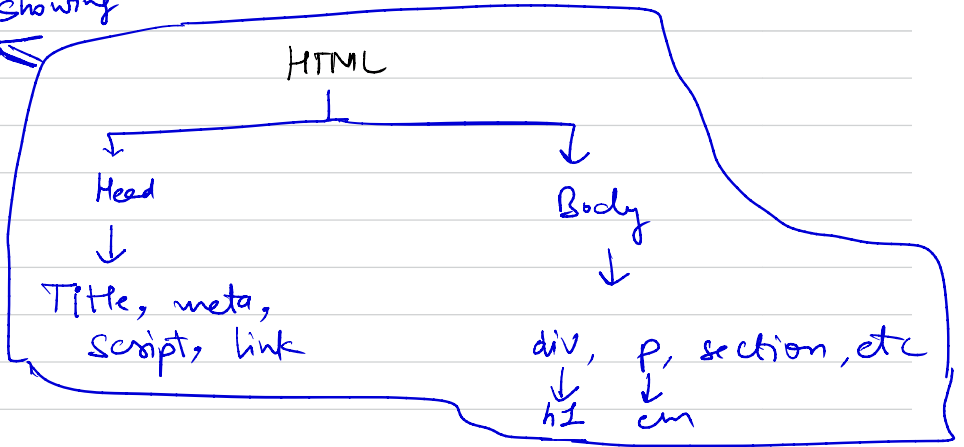
agar next line mei fir se h1 ko copy
karke likhoge, then h3 par class 3
apply ho jayega.

DOM TREE :-

(27:33)

this is showing

parent
child
relation



HTML ka children → head & body

Head & Body ke children hai → Title, meta, etc
div, p, etc

div ka bhai/sibling → p and section

sibling ke liye same parent hona chahiye.

Body & em ka kya relation hai? Body is ancestor of em

h1 and section mei koi relation nahi hai.

div is descendant of body.

```
<section>
```

```
  <article>
```

```
    <div>
```

```
      <h1>
```

```
        <p>sam</p>
```

```
        <h5>heading</h5>
```

```
      </h1>
```

```
    </div>
```

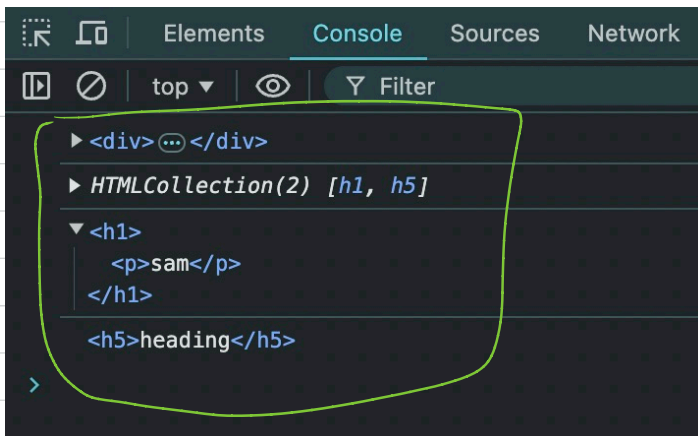
```
  </article>
```

```
</section>
```

→ index.
html

Ab mai `<div>` ki madad se `<p>` ki value nikalna chahata hu, but using JS. Kaise hoga?

```
let divv = document.querySelector('div');  
console.log(divv);  
console.log(divv.children);  
console.log(divv.children[0]);  
console.log(divv.children[1]);
```



yaha hum parent se child ke pass gaye,

Ab mai child se parent ke paas jama chahitahu.

```
let divv = document.querySelector('div');  
  
console.log(divv.parentElement);  
console.log(divv.parentElement.parentElement);  
console.log(divv.parentElement.parentElement.parentElement);  
console.log(divv.parentElement.parentElement.parentElement.parentElement);  
console.log(divv.parentElement.parentElement.parentElement.parentElement.parentElement);
```

```
▶ article ← app.js:17  
▶ section ← app.js:18  
▶ body ← app.js:19  
▶ html ← app.js:20  
null app.js:21  
> ↑
```

Html ka koi parent nahi hai → isliye null aaya.

```
let art = document.querySelector('article');  
  
console.log(art);  
console.log(art.children);  
console.log(art.children[0].children[0].children[0].children);
```

↪ 12

```
▶ <article>...</article> ←  
▶ HTMLCollection [div] ← → 11  
▶ HTMLCollection [] ← → 12  
>
```

l1 par hamne array mila jabki <article> ke andar sirf ek hi <div> hai. It does not depend on that ki kitne <div> hai. it always give an array.

In code $\Rightarrow$ l2 par hamne [0] square bracket use kiya, kyunki vo array hai.

l2 $\rightarrow$ empty array kyunki <p> ka koi children nahi hai.

(44:00)

next Element Sibling & previous Element Sibling:-

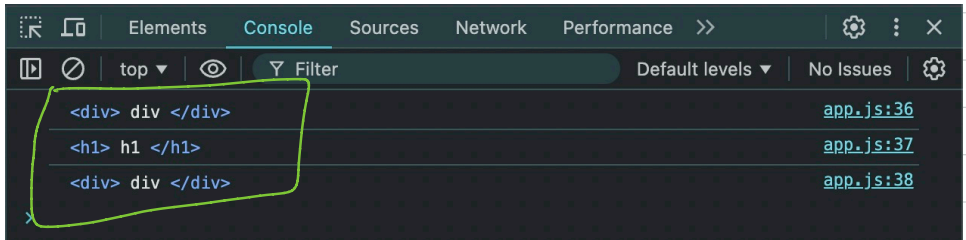
```
<body>
  <section>
    <article>
      art
    </article>
    <div>
      div
    </div>
    <h1>
      h1
    </h1>
    <p>sam</p>
  </section>
  <script src="app.js"></script>
</body>
```

```
let art = document.querySelector('article');
```

```
console.log(art.nextElementSibling);
```

```
console.log(art.nextElementSibling.nextElementSibling);
```

```
console.log(art.nextElementSibling.nextElementSibling.previousElementSibling);
```



Q Kya hum DOM ki madad se new element bana sakte hai? Yes, bana sakte hai. But kaise?

Creating New Element :-

```
let h2 = document.createElement('h2');
```

```
let p = document.createElement('p');
```

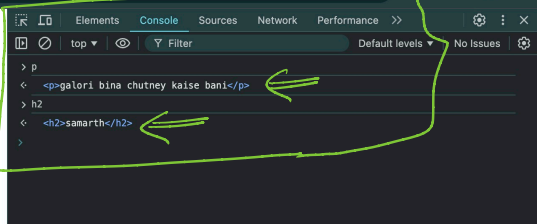
```
h2.innerText = 'samarth'
```

```
p.innerText = 'galori bina chutney kaise bani'
```

→ app.js

127.0.0.1:5500/creating-new-element/index.html

creating new element



```
<body>
```

```
<h1>creating new element</h1>
```

```
<script src="./app.js"></script>  
</body>
```

→ index.html

In the ss attached above, we can see ki hamare create kiye hue elements (p, h2) web page par display nahi hue. Now how to add them in your document or DOM.

DOM mei elements ko add karane ke liye, we have two things → ① appendChild()

② append()

```
<body>
```

```
<h1>creating new element</h1>
```

```
<section>
```

```
</section>
```

```
<script src="./app.js"></script>  
</body>
```

→ index.html

```
let section = document.querySelector('section');
```

```
let h2 = document.createElement('h2');
```

```
let p = document.createElement('p');
```

```
h2.innerText = 'samarth ka saamarth';
```

```
p.innerText = 'galori bina chutney kaise bani';
```

```
section.appendChild(h2);
```

→ l1

→ l2

l1 par hamne section tag ko select kiya, then usmei h2 ko add kar diya.

`appendChild ( h2 , p )` → ismei aap sirf ek hi element daal sakte ho.
↳ wrong syntax

Agar multiple elements daalne hai, then use karo `append()`.

mai `appendChild()` mei string pass nahi kar sakta to inserting any text into selected element.

`appendChild (" Hi I am Singh");`
↳ wrong syntax → you cant pass text/string.

(1:00:42)

```
let section = document.querySelector('section');  
let h2 = document.createElement('h2');  
let p = document.createElement('p');  
  
h2.innerText = 'samarth ka saamarth';  
p.innerText = 'galori bina chutney kaise bani';  
  
section.appendChild(h2); → l1  
section.appendChild(h2, p); → l2  
section.append('hi there all good?'); → l3
```

in case of `append()`, both statements will work l1 and l2.

(l2) multiple elements ek saath add karane hai, then you can use `.append ( h2, p );`

(Q3) Hum `append()` mei string bhi daal sakte hai.

creating new element

samarth ka saamarth

galori bina chutney kaise bani

hi there all good? → l3 o/p

→ l2 o/p

`append()` does not understand html tags.



← → ↻ 127.0.0.1:5500/creating-new-element/index.html

creating new element

samarth ka saamarth

galori bina chutney kaise bani

hi there all good?<h1>i am good!</h1>

```
let section = document.querySelector('section');
```

```
let h2 = document.createElement('h2');
```

```
let p = document.createElement('p');
```

```
h2.innerText = 'samarth ka saamarth';
```

```
p.innerText = 'galori bina chutney kaise bani';
```

```
section.append(h2, p);
```

```
section.append('hi there all good?');
```

```
section.append('<h1>i am good!</h1>');
```

```
section.append(h2, p, '<h1>i am good!</h1>');
```

→ you can also write like this. O/p same rahega.

Jis sequence mei likhoge usi sequence mei DOM ke andar jayega. pehle h2 jayega, then p jayega, then <h1>... </h1>.

agar mujhe <h1> ... </h1> pehle likhna hai,
then mai kaise karunga? we use

• `prepend()`

`section.append(p);`
`section.prepend(p);` } → you can't do this
ya to append()

```
let section = document.querySelector('section');  
  
let h2 = document.createElement('h2');  
let p = document.createElement('p');  
  
h2.innerText = 'samarth ka saamarth';  
p.innerText = 'galori bina chutney kaise bani';  
  
section.append(h2); ←  
section.append(p);  
section.prepend(h2); ←
```

Kara lo ya
prepend() Kara lo.
Both can't be done
on same element
together.

creating new element

samarth ka saamarth → h2

galori bina chutney kaise bani → p

prepend mei jo bhi likhoge wo ^{sabse} upar aayega.

append mei jo bhi likhoge, wo sabse neeche aayega.

The `prepend()` method inserts specified content at the beginning of the selected elements.

Tip: To insert content at the end of the selected elements, use the `append()` method.

```
let section = document.querySelector('section');
```

```
let h2 = document.createElement('h2');
```

```
let p = document.createElement('p');
```

```
h2.innerText = 'samarth ka saamarth';
```

```
p.innerText = 'galori bina chutney kaise bani';
```

```
section.prepend(h2, p, 'i am fine');
```

creating new element

samarth ka saamarth

galori bina chutney kaise bani

i am fine

`section.before(h2);` → agar h2 ko section ke
pehle append karana ho.

`section.after(h2)` → agar section ke band append
karana ho.

```
let section = document.querySelector('section');
```

```
let h2 = document.createElement('h2');
```

```
let p = document.createElement('p');
```

```
h2.innerText = 'samarth ka saamarth';
```

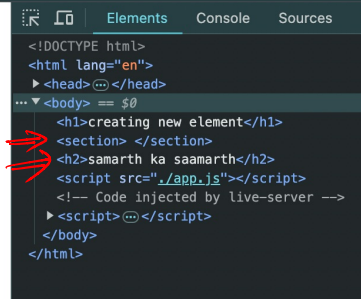
```
p.innerText = 'galori bina chutney kaise bani';
```

```
section.before(h2); → 1
```

```
section.after(h2); → 2
```

creating new element

samarth ka saamarth



```
<!DOCTYPE html>
<html lang="en">
  <head> </head>
  <body> == $0
    <h1>creating new element</h1>
    <section> </section>
    <h2>samarth ka saamarth</h2>
    <script src="/app.js"></script>
    <!-- Code injected by live-server -->
  </body>
</html>
```

`<section>` ke baad `<h2>` insert / append ho gya.

l1 and l2 mei se l2 ka effect ~~old~~ o/p mei dikhega kyunki l2 ~~me~~ ne l1 ko override kar liya.

Removing from DOM :-

`h2.remove()` → ek baar mei ek element hi remove hoga.

↳ hamne h2 ko select kiya, then h2 ko remove kar diya.

`div.removeChild(h2)` → agar selected element ka child remove karna ho.

```
<body>
```

```
<p>removing elements from dom</p>
```

```
<div>
```

```
<h1>hello 1</h1>
```

```
<h2>hello 2</h2>
```

```
<h3>hello 3</h3>
```

```
</div>
```

```
<script src="app.js"></script>
```

```
</body>
```

→ index.html

```
let h2 = document.querySelector('h2');
```

```
let h3 = document.querySelector('h3');
```

```
let div = document.querySelector('div');
```

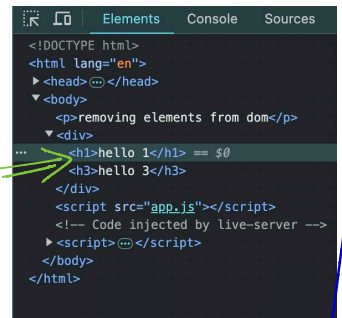
```
h2.remove();
```

→ app.js

removing elements from dom

hello 1

hello 3



```
<!DOCTYPE html>
<html lang="en">
  <head>
    <script src="app.js"></script>
  </head>
  <body>
    <p>removing elements from dom</p>
    <div>
      <h1>hello 1</h1>
      <h2>hello 2</h2>
      <h3>hello 3</h3>
    </div>
  </body>
</html>
```

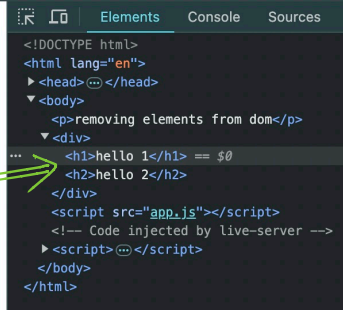
h2 remove ho gya.

```
let h2 = document.querySelector('h2');  
let h3 = document.querySelector('h3');  
let div = document.querySelector('div');  
div.removeChild(h3);
```

removing elements from dom

hello 1

hello 2



```
<!DOCTYPE html>  
<html lang="en">  
  <head></head>  
  <body>  
    <p>removing elements from dom</p>  
    <div>  
      <h1>hello 1</h1>  
      <h2>hello 2</h2>  
    </div>  
    <script src="app.js"></script>  
    <!-- Code injected by live-server -->  
  </body>  
</html>
```

div.removeChild (h2, h3) → this will not work. only one element can be removed at a time.

(1:36:36) Break over

#Events ⇒ Events matlab actions that happen on the document.

Important: Refer Shreyance sir notes to learn about events.

(Lec 20 3 Aug , 2024)

↳ you can find it inside current JS folder (directory).

Events are things that happen in the system you are programming, which the system tells you about so your code can react to them.

For example, if the user clicks a button on a webpage, you might want to react to that action by displaying an information box. In this article, we discuss some

Events are of almost infinite types (not actually infinite but
bahut se hote hai)

We can listen to any event in 3 ways:-

① inline way:-

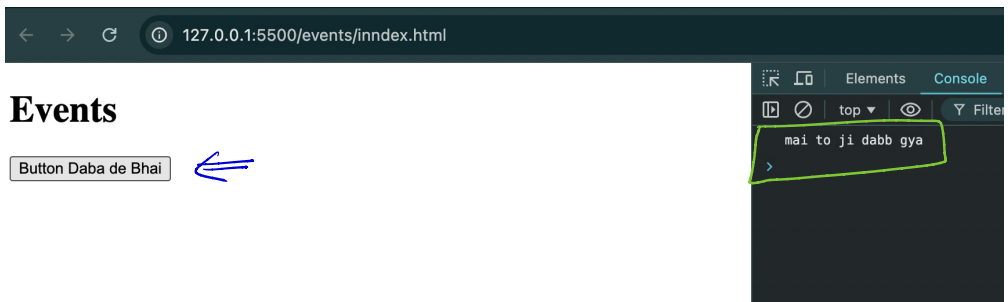
```
<h1>Events</h1>
```

```
<button onclick="something()" >Button Daba de Bhai</button>
```

→ index.html

```
function something(){  
  console.log("mai to ji dabb gya");  
}
```

→ app.js



button ke ander hamne "onclick" event add kiya tha.

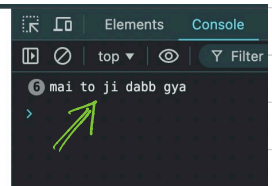
mere button ke click karne par mera event trigger ho raha hai.

Ques Will it work if I click on div?

```
<h1>Events</h1>
<div onclick="something()" >Button Daba de Bhai</div>
```

Events

Button Daba de Bhai



Note:- aap event kisi bhi div par laga sakte ho, jaise hamne yaha div ke upar lagaya.

onclick sab par lagta hai (naake sirf button par)

② 2nd way of creating event:-

```
let btn = document.querySelector('button');
btn.onclick = function(){
  console.log('jai bajrang bali');
};
```

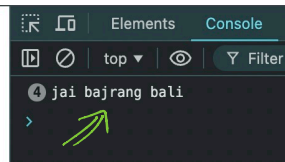
⇒ app.js

```
<button>Click me for getting jobs</button>
```

→ index.html

Events

Click me for getting jobs



```
let btn = document.querySelector('button');  
  
let samarth = function(){  
    console.log("bajrang bali ki jai")  
}  
  
btn.onclick = samarth;
```

↳ you can also write the last code like this.

Code:

javascript

Copy code

```
btn.onclick = samarth;
```

Execution:

- The `onclick` property of the `btn` element (the `<button>` in the HTML) is set to the `samarth` function.
- This means that when the button is clicked, the `samarth` function will be executed.

3. What Happens When You Click the Button?

- When you click the button, the `onclick` event is triggered.
- The `samarth` function is called because `btn.onclick` was assigned to it.
- Inside the `samarth` function, `console.log("bajrang bali ki jai")` is executed.
- The text `"bajrang bali ki jai"` appears in the browser's developer console.

4. Key Concepts in the Code

1. Event Handling (`onclick`):

- The `onclick` property is a way to attach a click event listener to an element.
- In this case, the `samarth` function is assigned to handle the click event.

2. Function Reference:

- The `samarth` function is not called directly with parentheses (`samarth()`).
- Instead, its reference (`samarth`) is passed to the `onclick` property, meaning the function will be invoked only when the event occurs (button click).

3. Console Logging:

- `console.log` is used for debugging and displaying information in the browser's developer console.

5. Why Use `btn.onclick` Instead of `onclick="..."` in HTML?

- Using `btn.onclick` in JavaScript keeps your HTML clean and separates structure (HTML) from behavior (JavaScript).
- It also makes your code easier to maintain and follow modern best practices.

Q guess the output :-

```
<button>naacho aur gaao</button>
```

→ index.html

```
let btn = document.querySelector('button');

function naacho(){
  console.log("natu natu natuuu...")
}

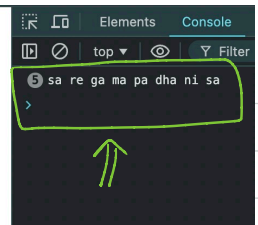
function gaayo(){
  console.log("sa re ga ma pa dha ni sa")
}

btn.onclick = naacho;
btn.onclick = gaayo;
```

→ app.js

Events

naacho aur gaao



btn.onClick ki value override ho gyi.
Is tarike se hum ek event par ek se jyada kaam
nahi kar sakte.

③ Third way :-

↳ we use it jab hame multiple
kaam ek saath karane ho.

addEventListener

Syntax :-

```
let btn = document.querySelector('button');  
btn.addEventListener('click', function(){  
  console.log("event chal gya")  
})
```

you can see ki hamne "click" use kiya instead
of "on click".

```

let btn = document.querySelector('button');

function naacho(){
    console.log("natu natu natuuu...")
}

function gaayo(){
    console.log("sa re ga ma pa dha ni sa")
}

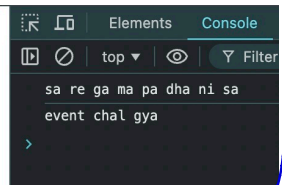
btn.onclick = naacho;
btn.onclick = gaayo;    → l1

btn.addEventListener('click', function(){    → l2
    console.log("event chal gya")
})

```

Events

naacho aur gaao



Both l1 and l2 worked.

2. Behavior of `onclick`

- The `onclick` property can hold only one function.
- When you assign a second function to `btn.onclick`, it overwrites the first one.

Step-by-step execution of your `onclick` code:

1. `btn.onclick = naacho;` sets the `onclick` property to the `naacho` function.
2. `btn.onclick = gaayo;` **overwrites** the `naacho` function with the `gaayo` function.

When you click the button, the `onclick` property now only calls the `gaayo` function.

3. Behavior of `addEventListener`

- The `addEventListener` method allows you to add multiple event listeners to the same event.
- Unlike `onclick`, it does not overwrite previous listeners—it just adds another one to the list.

Step-by-step execution of your `addEventListener` code:

1. `btn.addEventListener('click', function() { ... });` adds a new `click` event listener alongside the `onclick` property.

When you click the button, the following happens:

- The function assigned to `onclick` (i.e., `gaayo`) is executed.
- The additional `click` event listener added via `addEventListener` is also executed.

4. Why Are Both `gaayo` and `addEventListener` Working?

1. The `onclick` property executes the `gaayo` function because that's the last function assigned to it.
2. The `addEventListener` method ensures that the anonymous function (`console.log("event chal gya")`) is also executed because it adds a separate, independent event listener.

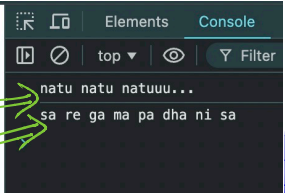
Q Ab mujhe nacho and gaayo dono function chalane hai. Kaise karu?

```
let btn = document.querySelector('button');  
  
function naacho(){  
    console.log("natu natu natuuu...")  
}  
  
function gaayo(){  
    console.log("sa re ga ma pa dha ni sa")  
}  
  
btn.addEventListener('click', naacho);  
btn.addEventListener('click', gaayo);
```

Hamne yaha ek
click se multiple
functions trigger
kardiye

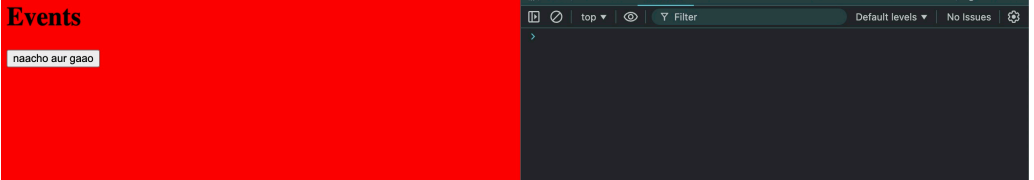
Events

naacho aur gaao



Button ke click karke hi background color of your document should be red.

```
let btn = document.querySelector('button');  
let body = document.querySelector('body');  
btn.addEventListener('click', function(){  
    body.style.backgroundColor = 'red';  
})
```



Technique → jo kaam karna hai use select
karo → button and body ko select
karo → then button par event
add kar do.

